



ROBOT EMBEDDED PROGRAMING

Project: Autonomous Delivery Mobile Robot



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1. Project Overview

The purpose of this project is to design, simulate, and build an autonomous mobile robot capable of carrying out delivery tasks. The robot must start at a designated location, navigate an environment that may contain randomly appearing obstacles, and reach a specified delivery point without direct human control. The task must be demonstrated in both a simulation environment and a physical prototype.

The robot design must adhere to basic constraints:

- Maximum dimensions: 300 mm × 200 mm × 200 mm
- Payload capacity: up to 100 grams

Students must make key engineering decisions about the robot's mechanical design, motion models, control systems, and navigation strategy.

2. Learning Objectives

- Apply kinematics and inverse kinematics to model and control robotic mobility.
- Implement motion planning to generate feasible paths under constraints.
- Demonstrate obstacle avoidance and adaptive control in dynamic environments.
- Integrate simulation results with the design and construction of a real prototype.
- Analyze and compare the performance of simulation vs. real implementation.
- Communicate design choices, testing methodology, and results in a professional engineering report.

3. Scope of Work

Teams will complete the following high-level tasks:

1. Conceptual Design: Define the design problem, identify constraints, and propose alternative solutions.
2. Simulation: Develop a functional ROS 2 simulation of the delivery robot that demonstrates motion from start to goal with obstacle avoidance.
3. Physical Build: Construct a real prototype capable of carrying out the same task within the given constraints.
4. Testing & Validation: Develop a testing plan, conduct trials in both simulation and real environments, and evaluate results.
5. Communication: Deliver a technical report and a live or recorded demo of the system.

4. Applications

- Healthcare: Hospital supply delivery (medications, lab samples, meals).
- Retail & Logistics: Last-mile delivery of goods and packages.

- Smart Cities & Campuses: Food delivery, document transport, and courier tasks.
- Industrial Facilities: Moving tools, components, or parts between workstations.

5. Equity, Diversity, and Inclusion (EDI) Considerations

Students are expected to reflect on how EDI principles apply to the design and use of delivery robots:

- Accessibility: How can autonomous delivery robots improve access to essential goods for people with mobility challenges or disabilities?
- Diverse Users: Consider cultural, linguistic, or interface needs to ensure inclusivity in human-robot interaction.
- Workforce Impact: Analyze how automation may affect employment and propose strategies to support displaced workers.
- Ethical Design: Address safety, privacy, and ethical concerns in deploying delivery robots in public or private spaces.

In the final report, teams must include a section on how EDI considerations influenced their design choices and their view of real-world applications.

6. Constraints and Assumptions

- Maximum size: 300 × 200 × 200 mm
- Payload capacity: up to 100 g
- Robot must operate in an indoor environment for the demo
- Obstacle locations will not be pre-programmed; the robot must adjust in real time
- Simulation must be done in ROS 2 (any distribution available in the lab)

7. Deliverables

1. Simulation: A working demonstration in ROS 2 showing navigation from start to delivery point.
2. Physical Build: A prototype capable of completing the same delivery task under constraints.
3. Final Report: A professional engineering document that includes:
 - Problem definition and scope
 - Design alternatives and justification of final design
 - Modeling, motion planning, and control approaches
 - Simulation setup and results
 - Physical build design, testing, and validation results
 - Applications and EDI considerations
 - Reflection on limitations and future improvements
4. Demo: A live or recorded demonstration of the robot in action.

8. Evaluation Rubric

Category	Criteria	Excellent (4)	Good (3)	Satisfactory (2)	Poor (1)
Design Process & Engineering Principles	Application of kinematics, inverse kinematics, motion planning, obstacle avoidance, and system modeling	Clear, correct, and rigorous application of all concepts; strong justification of design choices	Correct application with some gaps in justification or detail	Basic application with limited depth or accuracy	Minimal or incorrect application; weak or missing justification
Conceptual Design & Innovation	Creativity, feasibility, and consideration of constraints (mechanical, operational, environmental)	Innovative and well thought-out design with strong consideration of real-world constraints	Sound design with some creative elements and moderate constraints considered	Adequate design with minimal innovation or shallow consideration of constraints	Little creativity, weak feasibility, or missing constraint analysis
Simulation Implementation	Quality of ROS 2 simulation model, demonstration of delivery task	Fully functional simulation; accurate modeling and smooth task completion	Simulation works with minor errors or limited scope	Partial simulation with significant limitations	Incomplete or non-functional simulation
Physical Build & Integration	Functionality of real robot prototype, integration of subsystems	Robot consistently completes delivery task; strong mechanical and control integration	Robot completes task with minor issues or inconsistencies	Robot demonstrates partial task completion or limited integration	Robot fails to demonstrate meaningful functionality
Performance (Task Execution)	How effectively the robot carries out the delivery	Robot completes delivery task accurately,	Robot completes delivery task with minor	Robot completes task partially or with	Robot fails to complete the delivery task

	(speed, accuracy, reliability, repeatability)	efficiently, and consistently; meets all performance expectations	delays, inaccuracies, or inconsistencies	significant inefficiencies	
Testing & Validation	Systematic testing approach, comparison of simulation vs. real results	Thorough testing plan with strong analysis of performance and limitations	Good testing effort with moderate analysis of results	Basic testing with limited evaluation of outcomes	Little to no testing or evaluation provided
Communication & Final Report	Clarity, technical depth, structure, and reflection	Report is clear, well-organized, and demonstrates deep technical understanding; strong reflection on results and improvements	Report is clear and complete with good technical detail and reflection	Report is adequate but lacks depth, organization, or reflection	Report is incomplete, unclear, or lacks technical rigor
Teamwork & Professionalism	Collaboration, distribution of tasks, project management	Excellent teamwork with balanced contributions and professional conduct	Good teamwork with some uneven contributions	Adequate teamwork with limited collaboration	Poor teamwork, lack of coordination, or professionalism issues
Lifelong Learning	Evidence of independent learning, exploration of new tools, methods, or knowledge beyond class content	Team shows strong initiative, learning new tools/concepts independently and integrating them effectively	Team demonstrates some initiative in exploring new methods/tools and applying them with moderate success	Limited independent learning beyond provided material	No evidence of independent learning; relies only on provided material

Equity, Diversity, and Inclusion (EDI)	Consideration of EDI in design, applications, and impacts	Explicit integration of EDI considerations into design and reflection; strong awareness of societal impacts	Some integration of EDI considerations with moderate reflection	Limited mention of EDI with shallow reflection	No evidence of EDI considerations in design or analysis
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